

3.1: Definitions & Notation:

A sequence is a set of values, linked by a common rule.
A Series is defined as the sum of terms within a sequence.

Sigma Notation: Sum of a sequence

$\sum_{n=1}^{10} S_n$: Sums terms in Sequence S from the 1st to 10th Term.

$\sum_{n=a}^{b} S_n$: Series from a^{th} to b^{th}.

Deductive Sequences define an nth term within a sequence, whereas an inductive sequence links the nth term to the $(n-1)^{th}$ term. Inductive sequences must therefore define the first term.

Sequences may converge to a number as the difference between terms decreases, it may diverge as the difference. A sequence is periodic if it repeats itself at regular intervals

Monotonicity:

A sequence is increasing iff $\forall n \in \mathbb{N}$ $a_{n+1} \geq a_n$
A sequence is decreasing iff $\forall n \in \mathbb{N}$ $a_{n+1} \leq a_n$
} Unstrict Definitions

A sequence is only monotonic iff it is decreasing or increasing.

3.2: Arithmetic Sequences & Series:

A sequence in which consecutive terms differ by the addition of a constant. This constant is the common difference

Therefore, it is defined by: $S_n = a + (n-1)d$, where a is the first term, and n is the position in the sequence ($\mathbb{N}$).
$d = u_{n+1} - u_n$
The series of this sequence can be seen as:
The average of all terms is the middle term, also: $\frac{(a+l)}{2}$, where l is the last term.

Hence: $S_n = \frac{n(a+l)}{2} = \frac{n}{2}(2a + (n-1)d)$. This is always 1 based indexing

3.3: Geometric Sequences & Series:

A sequence in which consecutive terms differ by the multiplication of a constant. This constant is the common ratio.

Therefore, it is defined by: $S_n = ar^{n-1}$, where a is the 1st term. $n \in \mathbb{N}$ $\quad r = \frac{S_{n+1}}{S_n}$

$$\sum_{n=1}^{N} ar^{n-1} = a + ar + ar^2 + \dots + ar^{N-1} = S_N$$

$$rS_N + a = S_N + ar^N$$
$$\Rightarrow S_N(r-1) = a(r^N - 1)$$
$$\Rightarrow S_N = \frac{a(r^N-1)}{r-1} = \frac{a(1-r^N)}{1-r} \quad | \; n \in \mathbb{N}$$

An infinite geometric series only converges given $|r|<1$, so difference in terms decrease.

$$\Rightarrow S_\infty = \lim_{N\to\infty} \frac{a(1-r^N)}{1-r} = \frac{a}{1-r} \quad | \; |r|<1$$